

CIFAR-10 Dataset Explained

Item	Description
Full Form	Canadian Institute For Advanced Research - 10 Classes
Type	Image Classification Dataset
Total Images	60,000 Images
Training Images	50,000 Images
Testing Images	10,000 Images
Number of Classes	10
Image Size	32 × 32 Pixels

Channels RGB (3 Channels)

Common CNN, Deep Learning,
Use Computer Vision Practice

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CIFAR-10 Classes

Label	Class
0	Airplane 
1	Automobile 
2	Bird 
3	Cat 
4	Deer 

- 5 Dog 🐕
- 6 Frog 🐸
- 7 Horse 🐎
- 8 Ship 🚢
- 9 Truck 🚚

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Dataset Structure

Dataset	Number of Images
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Training Set	50,000
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Test Set 10,000

Total 60,000

Each class contains:

$50,000 \div 10 = 5,000$ training images

$10,000 \div 10 = 1,000$ testing images

Example Images

Class	Example Image
Airplane	Aircraft in sky
Automobile	Car
Bird	Sparrow, Parrot
Cat	Cat
Deer	Deer
Dog	Dog

Frog	Frog
Horse	Horse
Ship	Boat, Ship
Truck	Lorry, Truck

Why CIFAR-10 is Popular?

Reason	Explanation
Small Dataset	Easy to download and train
Standard Benchmark	Used worldwide to compare CNN models
Multi-Class Problem	Contains 10 different categories
Good for Learning	Best dataset for CNN beginners
Fast Training	Images are only 32×32 pixels

In Your Code

```
trainset = torchvision.datasets.CIFAR10(  
    root='./data',  
    train=True,  
    download=True,  
    transform=transform  
)
```

Meaning

Parameter

Purpose

`root='./data'`

Dataset storage
folder

`train=True`

Load training data

`download=True`

Download
automatically if not
present

`transform=transform`

Apply
preprocessing

Data Flow

CIFAR-10 Images



Preprocessing



CNN Model



Feature Extraction



Classification



10 Classes

Interview Answer

Question

What is CIFAR-10?

Answer

CIFAR-10 is a popular image classification dataset containing 60,000 color images across 10 classes, commonly used to train and evaluate CNN models.

Why is it used?	To learn and test image classification algorithms and CNN architectures.
How many classes are there?	10 classes.
What is the image size?	32×32 RGB images.

One-Line Summary

CIFAR-10 is a benchmark image dataset containing 60,000 small RGB images divided into 10 object categories, widely used for CNN and deep learning image classification projects.